



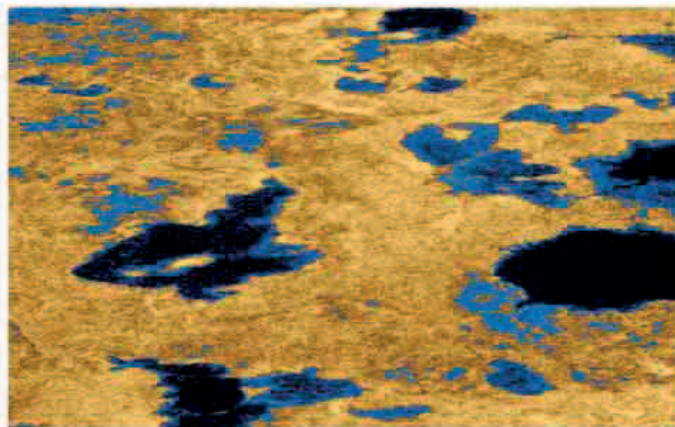
“ A NEW PAGE OF AERONAUTICAL HISTORY HAS BEEN WRITTEN. ”

Jacques Chirac, French politician, 2005

The Airbus A380 takes off from its production site in Toulouse, France, on its maiden flight on 27 April 2005. The world's largest passenger airliner, the A380 can carry up to 853 people.

IN THE FIRST-EVER LANDING OF A SPACECRAFT in the outer Solar System—and only the second landing on any moon—the European Space Agency's (ESA's) Huygens probe parachuted on to **the surface of Saturn's largest moon, Titan**, in January. The probe was part of the joint NASA-ESA Cassini-Huygens mission, and the spacecraft that carried Huygens to Titan, Cassini, remained in orbit around Titan, gathering data and images and relaying signals from Huygens back to

astronomers on Earth. Titan had long been of interest to astrobiologists, as its **atmosphere was known to contain organic compounds**—carbon-rich compounds that could form the basis of life (see panel, below). The Huygens probe captured more than 300 images of Titan's surface during its descent. Images of the surface taken after landing show rock-shaped objects on “sandy” ground—although these objects and the sand are both probably mostly frozen water.



LIFE ON TITAN?

Titan's atmosphere has clouds that produce rain of the carbon compound methane, and the space probe Cassini found lakes of methane and ethane (the blue areas in this color-enhanced radar image) around Titan's poles. In 2012, NASA also found evidence of a huge subsurface ocean of water, which led to speculation that simple life forms may exist on this seemingly inhospitable world.

In the following month, an international team of astronomers released their analysis of a **huge, galaxy-sized mass of hydrogen** known as **VIRGOHI21**, which lies about 50 million light years away and had been discovered by radio astronomers in 2004. The motion of the galaxy suggests that it **may be composed mostly of cold, dark matter**—a strange form of matter that does not interact with light or other types of electromagnetic radiation but does have a gravitational effect (see 2003). Current astrophysical theories suggest that all galaxies contain dark matter—it is the only way to account for the motion and distribution of the galaxies' stars. However, VIRGOHI21 was the first strong candidate for a dark matter galaxy, a galaxy dominated by dark matter.

In March, US paleontologist **Mary Schweitzer** reported finding **68-million-year-old soft tissue**, which she had extracted from the fossilized femur (thigh bone) of a ***Tyrannosaurus rex*** dinosaur by dissolving away the mineral parts of the fossil. This method is commonly used on living bone as a way of extracting soft tissue. The same procedure carried out on a fossil would normally dissolve the whole specimen, since most fossils

are completely mineralized. However, with this specimen the process revealed a soft, elastic bone matrix made mostly of what seemed like collagen. Viewing this matrix under a microscope, Schweitzer found what appeared to be **blood vessels and bone cells**. She saw similar structures in soft tissue extracted from the femur of a modern ostrich, a close relative of *T. rex*. Repeating the procedure on other dinosaur fossils, Schweitzer found two other samples of soft tissue. Many scientists were sceptical of the analysis, suggesting that the organic matter was the result of contamination, but more detailed examination in 2008 and 2011 seemed to support Schweitzer's interpretation.

In April, a new aircraft made its maiden flight: the **Airbus A380**, which replaced the Boeing 747



Facing the press

The first person to receive a partial face transplant, Isabelle Dinoire gave a press conference in February 2006, only three months after her pioneering operation.

(see 1970) as the **world's largest commercial passenger aircraft**. Designed and built by European corporation Airbus, the wide-bodied A380 has two decks and is 239 ft (73 m) long. It began commercial service in 2007.

Toward the end of the year, in November, 38-year-old Frenchwoman Isabelle Dinoire became the **first person to have a partial face transplant**. She received skin, blood vessels, and muscles after her face had been badly damaged when she was mauled by a dog six months earlier. The first full face transplant was performed in Spain in 2010.

3,200 MILES THE DIAMETER OF TITAN, SATURN'S LARGEST MOON

January 5 Astronomers discover **Eris**, a rocky body larger than Pluto orbiting beyond Neptune

January 14 The Huygens probe lands on Saturn's moon **Titan**

February 23 Astronomers report that they have located a possible **dark matter** galaxy, **VIRGOHI21**

March 25 US paleontologist **Mary Schweitzer** reports the first recovery of soft tissue from a dinosaur

April 27 Maiden flight of **Airbus A380**, the world's largest passenger airliner

November 27 Surgeons in France perform the first partial face transplant

December 8 Scientists publish the completed sequence of the dog genome